

Ocular Microbiome Project Website Overview

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The **Ocular Microbiome Project** website is available at <https://www.ocular-microbiome.org>.

- It uses the [ocular-microbiome.org](https://www.ocular-microbiome.org) domain with secure SSL access and Google Analytics for traffic monitoring.
- [GitHub](#) version control with continuous integration and delivery (CI/CD) is used for backup, automated building and validation.
- The site is built using the [Sphinx](#) documentation generator and the [Read the Docs](#) hosting service and theme.
- Content is stored as simple **Markdown**, while data tables are stored in **CSV** format.
- The **HTML/CSS** web content is automatically generated by **Sphinx** whenever the site is rebuilt.
- This setup provides a clean, maintainable structure that makes updates and readability easy.
- The site is linked to JHU [SciServer](#), which is used to upload and store raw sequence files.
- Client-side JavaScript (jQuery) is used for interactive functionality.
- All non-numeric data is indexed, allowing users to search across the entire website.
- All tables support row and column filtering, sorting, and searching.

Main Page

The website's main page contains project, grant, consortium, and contact information. (launched in **July 2023** by the **National Eye Institute**, NIH grant U24-EY035078, Lisa A. Neuhold grant administrator).

A more detailed [Consortium Member](#) page lists all participating research centers, their Principal Investigators (PIs), and relevant contact information. (**Baylor College of Medicine; University of Washington; Vanderbilt University Medical Center; Johns Hopkins University; and University of Pittsburgh**).

Main Sections

The two main sections are [Consortium Data](#) and [Consortium Results](#).

Consortium Data

[Consortium Data](#) is intended to include information about the raw data generated by each center.

Entries for all five centers have been created; however, only the Johns Hopkins University section has been populated so far.

The [JHU Data](#) page includes three sections: **Libraries**, **Samples**, and **Files**.

- [Libraries](#) Contains a table with library processing dates, sample counts, and metadata (additional metadata to be added).
- [Samples](#) Includes a table with sample IDs, sample names, total numbers of read pairs, human reads, non-human reads, and additional metadata.
The table is searchable and sortable, and contains hyperlinks to non-human FASTQ.gz files stored on SciServer, which can be downloaded directly.
- [Files](#) All nonhuman read files generated by JHU up to date can also be downloaded in bulk from this section. The URL is: https://data.idies.jhu.edu/OcularMicrobiome/Consortium_data/JHU/
These files are in FASTA.gz format and contain the remaining reads after removing those aligned by Bowtie2 to the human GRCh38/T2T-CHM13 reference genomes or classified as human by KrakenUniq. A copy of the scripts used is available on [GitHub](#).

JHU Non-Human Data (Public on SciServer)

Date: 2025/08/05

Column Name	#Samples	Min	Median	Max	#Read-Pairs	#Bases	Gzipped
total_read_pairs	193	717,757	16,552,663	98,376,897	3,586,136,845	1,075,841,053,500	158G
non_human_read_pairs	193	4,579	1,513,206	97,247,641	1,749,204,017	524,761,205,100	
human_read_pairs	193	148,151	3,062,334	63,158,309	1,836,932,827	551,079,848,100	
kraken_assigned_human_read_pairs	193	87	5,977	52,367	1,284,406	385,321,800	

- Note: read_pairs=2*150bp

Consortium Results

[Consortium Results](#) is intended to include results generated by each group.

Entries for all five centers have been created; however, only the JHU section has been populated so far.

The [JHU Results](#) page lists the top species identified at the project, library, and sample levels.

Taxonomic names, IDs, read counts, and percentages are provided in searchable and sortable tables.

Previous Microbiome Studies

[Previous Microbiome Studies](#) & [Analysis of Previous Studies](#) includes data and results from 44 public projects available through NCBI.

These data have been downloaded and reanalyzed using the same methodology.

The Bulk non-human FASTA.gz files are available to download from

https://data.idies.jhu.edu/OcularMicrobiome/NCBI_Public_Data/

Metadata

[Metadata](#) contains a proposed organization of data by JHU, enabling samples and results to be linked and searched across the site.

It includes **Project**, **Center**, **Individual**, **BioSample**, **Library**, **Sample**, and **Analysis** tables, all connected by **id** fields.

So far we have generated a list of **seven core tables**:

- **Project** : Contains high-level metadata about each research initiative or BioProject.
- **Center** : Lists metadata about the participating research centers (e.g., university, lab).
- **Individual** : Captures participant-level information (e.g., subject IDs, demographics).
- **BioSample** : Describes the biological material itself (e.g., eye swabs), following a scheme similar to NCBI's BioSample model.
- **Library** : Records details about how sequencing libraries were prepared (e.g., protocols, dates, sample counts).
- **Sample** : Stores data about each sequencing sample, including links to both BioSamples and libraries, and associated metadata (e.g., read counts, QC, etc.).
- **Analysis** : Contains information about downstream analyses performed on the samples (e.g., taxonomic classification, read assignments), enabling users to trace from raw data to processed results.